

Prepare: Lewis §§3.1–3.3

Primary objective. Explain the different goals of thermal- and fast-spectrum reactor designs using $\eta(E)$, ξ , Σ_s , and Σ_a .

1 Energy and the multiplication factor

Recall the generation-to-generation definition:

$$k = \frac{\text{\# neutrons in generation } i}{\text{\# neutrons in generation } i-1}$$

What determines this value? Chapter 3 focuses on **energy**: neutron cross sections are strongly energy dependent.

The model for this chapter

Infinite Set aside spatial diffusion and leakage.
Homogeneous Use a spatially uniform composition.
Steady state Set aside time-dependent phenomena, including delayed-neutron dynamics.
 The notes write the multiplication factor as k_∞ .

Chapter roadmap. Fuels → moderators → energy distributions → energy averaging → k_∞ in terms of energy-averaged (“effective”) cross sections.

Energy range of interest. Roughly

$$0.001 \text{ eV} \lesssim E \lesssim 10 \text{ MeV}.$$

2 Fuel: neutrons produced per absorption

The energy-dependent ratio $\eta(E)$

$$\eta(E) = \frac{\nu(E) \Sigma_f(E)}{\Sigma_a(E)} = \frac{\text{fission neutrons produced}}{\text{neutrons absorbed}}.$$

$\nu(E)$ Mean neutrons produced per fission.
 $\Sigma_f(E)$ Macroscopic fission cross section.
 $\Sigma_a(E)$ Macroscopic absorption cross section.
 Use the same fuel composition in the numerator and denominator.

What do the two ratios count?

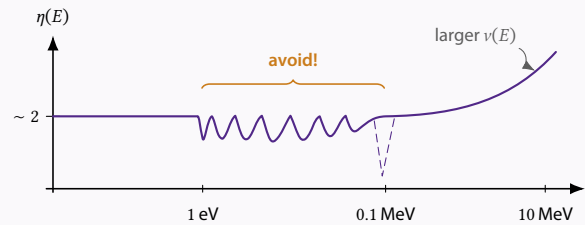
How does $\eta(E)$ differ from k_∞ ? Which compares generations, and which counts production per absorption at a specified energy?

Board work: absorption in a fuel mixture

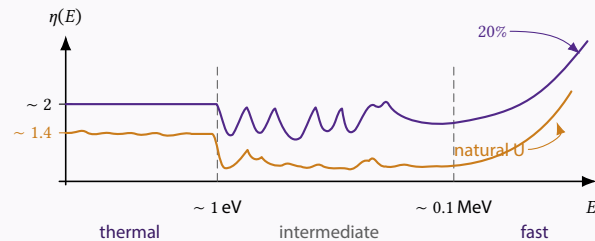
What nuclide densities and microscopic data are needed to construct $\Sigma_a(E)$ for a uranium mixture?

3 Read the fuel sketches

Pure fissile material



Natural uranium and an enriched mixture



Note, these are cartoons to illustrate the concepts!

Ponderable

If $\eta(E)$ represents an upper bound on k_∞ , why do you think we rarely see reactors operating in the intermediate spectrum?

4 Enrichment: How Much is Enough?

Naval reactors typically use very high enrichment, while civilian fuels range from natural uranium to 20% enrichment, the arbitrary(?) cutoff between low-enriched uranium **LEU** and **HEU**. Traditional power reactors typically use enrichments below 5%; enrichments between 5 and 20% are known as High-Assay LEU or HALEU.

Reading follow-up

Why might a design favor high enrichment? Why might a design favor low enrichment?

5 Thermal and fast design strategies

Thermal-spectrum reactors

Use **low-mass moderators** to move neutrons from fast to thermal energies with as **few collisions in the resonance range** as possible. The notes connect suitable fuel-moderator choices with low or no enrichment.

Fast-spectrum reactors

Choose materials that do not slow neutrons effectively: **avoid low-mass materials** and **minimize inelastic scattering**.

Keep the mechanisms distinct. The moderator measures below concern **elastic** scattering. At higher energies, **inelastic** scattering becomes significant, affecting fast spectra.

6 What makes a good moderator?

Three desirable properties

- Large ξ** A light target gives a large average logarithmic energy loss per elastic collision.
- Large Σ_s** Large number density and/or large σ_s favors collisions in the moderator.
- Small Σ_a** Prefer scattering to absorption; the notes emphasize $\sigma_a \ll \sigma_s$.

Review from Lesson 07: decrement and collision count

For elastic scattering isotropic in the CM frame, with target-to-neutron mass ratio A ,

$$\alpha = \frac{(A-1)^2}{(A+1)^2}, \quad \xi = 1 + \frac{\alpha \ln \alpha}{1-\alpha}.$$

At $A = 1$, use the limiting value $\xi = 1$.

$$n \approx \frac{1}{\xi} \ln \frac{E_0}{E_n}.$$

For compounds, use the effective mixture decrement; the Lesson 07 water example uses $\xi_{\text{water}} = 0.93$.

One favorable property is not enough

Which moderator in the table gives the fewest collisions for a specified E_0/E_n ? Does the same moderator have the largest slowing-down ratio?

Notes: connecting the three properties

7 Slowing-down power and ratio

Combine the moderator properties

$$\text{slowing-down power} = \xi \Sigma_s,$$

$$\text{slowing-down ratio} = \frac{\xi \Sigma_s}{\Sigma_a}.$$

The first combines energy loss and scattering; the second compares that combination with absorption.

Moderator comparison

	ξ	$\xi \Sigma_s$	$\xi \Sigma_s / \Sigma_a$
H ₂ O	0.93	1.28	58
D ₂ O	0.51	0.18	21,000
C	0.158	0.056	200

Source limitation. Values are reproduced as transcribed. The notes do not specify reference conditions or the units used for $\xi \Sigma_s$.

Natural-uranium connection. Heavy water and graphite are moderators with which natural-U fuel can be used; graphite was using in Fermi's pile.

8 Use the measures

Recover absorption from the table

Rearrange the slowing-down ratio to obtain

$$\Sigma_a = \frac{\xi \Sigma_s}{\text{slowing-down ratio}}.$$

Then compute the implied absorption cross sections:

$$\text{H}_2\text{O} : \Sigma_a = \frac{1.28}{58} \approx \text{_____},$$

$$\text{D}_2\text{O} : \Sigma_a = \frac{0.18}{21,000} \approx \text{_____}.$$

The results have the same inverse-length units as the table's slowing-down power; no numerical unit convention is supplied here.

Compare the design choices

Why can water have the largest slowing-down power in the table but the smallest slowing-down ratio? How does this connect to different fuel choices?